

which is in the quadratic form $\mathbf{x}^\top \mathbf{A} \mathbf{x} + \mathbf{b}^\top \mathbf{x} + c = 0$. Therefore, if we consider Gaussian distributions for the two classes, the decision boundary of classification is quadratic. Because of quadratic decision boundary which discriminates the two classes, this method is named *quadratic discriminant analysis*.

For obtaining Eq. (25), we brought the expressions to the right side which was corresponding to the second class; therefore, if we use $\delta(\mathbf{x}) : \mathbb{R}^d \rightarrow \mathbb{R}$ as the left-hand-side expression (function) in Eq. (25):

$$\begin{aligned} \delta(\mathbf{x}) := & \mathbf{x}^\top (\Sigma_1 - \Sigma_2)^{-1} \mathbf{x} + 2 (\Sigma_2^{-1} \mu_2 - \Sigma_1^{-1} \mu_1)^\top \mathbf{x} \\ & + (\mu_1^\top \Sigma_1^{-1} \mu_1 - \mu_2^\top \Sigma_2^{-1} \mu_2) + \ln\left(\frac{|\Sigma_1|}{|\Sigma_2|}\right) + 2 \ln\left(\frac{\pi_2}{\pi_1}\right), \end{aligned} \quad (26)$$

the class of an instance $\mathbf{x}$ is estimated as the Eq. (22).

If the priors of two classes are equal, i.e., $\pi_1 = \pi_2$, the Eq. (20) becomes:

$$\begin{aligned} & \mathbf{x}^\top (\Sigma_1 - \Sigma_2)^{-1} \mathbf{x} + 2 (\Sigma_2^{-1} \mu_2 - \Sigma_1^{-1} \mu_1)^\top \mathbf{x} \\ & + (\mu_1^\top \Sigma_1^{-1} \mu_1 - \mu_2^\top \Sigma_2^{-1} \mu_2) + \ln\left(\frac{|\Sigma_1|}{|\Sigma_2|}\right) = 0, \end{aligned} \quad (27)$$

whose left-hand-side expression can be considered as $\delta(\mathbf{x})$ in Eq. (22).

5. LDA and QDA for Multi-class Classification

Now we consider multiple classes, which can be more than two, indexed by $k \in \{1, \dots, |\mathcal{C}|\}$. Recall Eq. (12) or (15) where we are using the scaled posterior, i.e., $f_k(\mathbf{x}) \pi_k$. According to Eq. (16), we have:

$$\begin{aligned} & f_k(\mathbf{x}) \pi_k \\ &= \frac{1}{\sqrt{(2\pi)^d |\Sigma_k|}} \exp\left(-\frac{(\mathbf{x} - \mu_k)^\top \Sigma_k^{-1} (\mathbf{x} - \mu_k)}{2}\right) \pi_k. \end{aligned}$$

Taking natural logarithm gives:

$$\begin{aligned} \ln(f_k(\mathbf{x}) \pi_k) = & -\frac{d}{2} \ln(2\pi) - \frac{1}{2} \ln(|\Sigma_k|) \\ & - \frac{1}{2} (\mathbf{x} - \mu_k)^\top \Sigma_k^{-1} (\mathbf{x} - \mu_k) + \ln(\pi_k). \end{aligned}$$

We drop the constant term $-(d/2) \ln(2\pi)$ which is the same for all classes (note that this term is multiplied before taking the logarithm). Thus, the scaled posterior of the k -th class becomes:

$$\begin{aligned} \delta_k(\mathbf{x}) := & -\frac{1}{2} \ln(|\Sigma_k|) \\ & - \frac{1}{2} (\mathbf{x} - \mu_k)^\top \Sigma_k^{-1} (\mathbf{x} - \mu_k) + \ln(\pi_k). \end{aligned} \quad (28)$$

In QDA, the class of the instance $\mathbf{x}$ is estimated as:

$$\hat{\mathcal{C}}(\mathbf{x}) = \arg \max_k \delta_k(\mathbf{x}), \quad (29)$$

because it maximizes the posterior of that class. In this expression, $\delta(\mathbf{x})$ is Eq. (28).

In LDA, we assume that the covariance matrices of the k classes are equal:

$$\Sigma_1 = \dots = \Sigma_{|\mathcal{C}|} = \Sigma. \quad (30)$$

Therefore, the Eq. (28) becomes:

$$\begin{aligned} \delta_k(\mathbf{x}) = & -\frac{1}{2} \ln(|\Sigma|) \\ & - \frac{1}{2} (\mathbf{x} - \mu_k)^\top \Sigma^{-1} (\mathbf{x} - \mu_k) + \ln(\pi_k) = -\frac{1}{2} \ln(|\Sigma|) \\ & - \frac{1}{2} \mathbf{x}^\top \Sigma^{-1} \mathbf{x} - \frac{1}{2} \mu_k^\top \Sigma^{-1} \mu_k + \mu_k^\top \Sigma^{-1} \mathbf{x} + \ln(\pi_k). \end{aligned}$$

We drop the constant terms $-(1/2) \ln(|\Sigma|)$ and $-(1/2) \mathbf{x}^\top \Sigma^{-1} \mathbf{x}$ which are the same for all classes (note that before taking the logarithm, the term $-(1/2) \ln(|\Sigma|)$ is multiplied and the term $-(1/2) \mathbf{x}^\top \Sigma^{-1} \mathbf{x}$ is multiplied as an exponential term). Thus, the scaled posterior of the k -th class becomes:

$$\delta_k(\mathbf{x}) := \mu_k^\top \Sigma^{-1} \mathbf{x} - \frac{1}{2} \mu_k^\top \Sigma^{-1} \mu_k + \ln(\pi_k). \quad (31)$$

In LDA, the class of the instance $\mathbf{x}$ is determined by Eq. (29), where $\delta(\mathbf{x})$ is Eq. (31), because it maximizes the posterior of that class.

In conclusion, QDA and LDA deal with maximizing the *posterior* of classes but work with the *likelihoods* (class conditional) and *priors*.

6. Estimation of Parameters in LDA and QDA

In LDA and QDA, we have several parameters which are required in order to calculate the posteriors. These parameters are the means and the covariance matrices of classes and the priors of classes.

The priors of the classes are very tricky to calculate. It is somewhat a chicken and egg problem because we want to know the class probabilities (priors) to estimate the class of an instance but we do not have the priors and should estimate them. Usually, the prior of the k -th class is estimated according to the sample size of the k -th class:

$$\hat{\pi}_k = \frac{n_k}{n}, \quad (32)$$

where n_k and n are the number of training instances in the k -th class and in total, respectively. This estimation considers Bernoulli distribution for choosing every instance out of the overall training set to be in the k -th class.

The mean of the k -th class can be estimated using the Maximum Likelihood Estimation (MLE), or Method of Moments (MOM), for the mean of a Gaussian distribution:

$$\mathbb{R}^d \ni \hat{\mu}_k = \frac{1}{n_k} \sum_{i=1}^n \mathbf{x}_i \mathbb{I}(\mathcal{C}(\mathbf{x}_i) = k), \quad (33)$$

where $\mathbb{I}(\cdot)$ is the indicator function which is one and zero if its condition is satisfied and not satisfied, respectively.

In QDA, the covariance matrix of the k -th class is estimated using MLE:

$$\mathbb{R}^{d \times d} \ni \hat{\Sigma}_k = \frac{1}{n_k} \sum_{i=1}^n (\mathbf{x}_i - \hat{\boldsymbol{\mu}}_k)(\mathbf{x}_i - \hat{\boldsymbol{\mu}}_k)^\top \mathbb{I}(\mathcal{C}(\mathbf{x}_i) = k). \quad (34)$$

Or we can use the *unbiased* estimation of the covariance matrix:

$$\mathbb{R}^{d \times d} \ni \hat{\Sigma}_k = \frac{1}{n_k - 1} \sum_{i=1}^n (\mathbf{x}_i - \hat{\boldsymbol{\mu}}_k)(\mathbf{x}_i - \hat{\boldsymbol{\mu}}_k)^\top \mathbb{I}(\mathcal{C}(\mathbf{x}_i) = k). \quad (35)$$

In LDA, we assume that the covariance matrices of the classes are equal; therefore, we use the weighted average of the estimated covariance matrices as the common covariance matrix in LDA:

$$\mathbb{R}^{d \times d} \ni \hat{\Sigma} = \frac{\sum_{k=1}^{|C|} n_k \hat{\Sigma}_k}{\sum_{r=1}^{|C|} n_r} = \frac{\sum_{k=1}^{|C|} n_k \hat{\Sigma}_k}{n}, \quad (36)$$

where the weights are the cardinality of the classes.

7. LDA and QDA are Metric Learning!

Recall Eq. (28) which is the scaled posterior for the QDA. First, assume that the covariance matrices are all equal (as we have in LDA) and they all are the identity matrix:

$$\Sigma_1 = \dots = \Sigma_{|C|} = \mathbf{I}, \quad (37)$$

which means that all the classes are assumed to be spherically distributed in the d dimensional space. After this assumption, the Eq. (28) becomes:

$$\delta_k(\mathbf{x}) = -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_k)^\top (\mathbf{x} - \boldsymbol{\mu}_k) + \ln(\pi_k), \quad (38)$$

because $|\mathbf{I}| = 1$, $\ln(1) = 0$, and $\mathbf{I}^{-1} = \mathbf{I}$. If we assume that the priors are all equal, the term $\ln(\pi_k)$ is constant and can be dropped:

$$\delta_k(\mathbf{x}) = -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_k)^\top (\mathbf{x} - \boldsymbol{\mu}_k) = -\frac{1}{2}d_k^2, \quad (39)$$

where d_k is the Euclidean distance from the mean of the k -th class:

$$d_k = \|\mathbf{x} - \boldsymbol{\mu}_k\|_2 = \sqrt{(\mathbf{x} - \boldsymbol{\mu}_k)^\top (\mathbf{x} - \boldsymbol{\mu}_k)}. \quad (40)$$

Thus, the QDA or LDA reduce to simple Euclidean distance from the means of classes if the covariance matrices are all identity matrix and the priors are equal. Simple

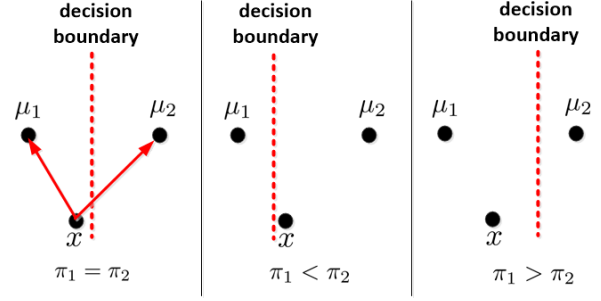


Figure 2. The QDA and LDA where the covariance matrices are identity matrix. For equal priors, the QDA and LDA reduce to simple classification using Euclidean distance from means of classes. Changing the prior modifies the location of decision boundary where even one point can be classified differently for different priors.

distance from the mean of classes is one of the simplest classification methods where the used metric is Euclidean distance.

The Eq. (39) has a very interesting message. We know that in metric Multi-Dimensional Scaling (MDS) (Cox & Cox, 2000) and kernel Principal Component Analysis (PCA), we have (see (Ham et al., 2004) and Chapter 2 in (Strange & Zwiggelaar, 2014)):

$$\mathbf{K} = -\frac{1}{2} \mathbf{H} \mathbf{D} \mathbf{H}, \quad (41)$$

where $\mathbf{D} \in \mathbb{R}^{n \times n}$ is the distance matrix whose elements are the distances between the data instances, $\mathbf{K} \in \mathbb{R}^{n \times n}$ is the kernel matrix over the data instances, $\mathbb{R}^{n \times n} \ni \mathbf{H} := \mathbf{I} - (1/n)\mathbf{1}\mathbf{1}^\top$ is the centering matrix, and $\mathbb{R}^n \ni \mathbf{1} := [1, 1, \dots, 1]^\top$. If the elements of the distance matrix $\mathbf{D}$ are obtained using Euclidean distance, the MDS is equivalent to Principal Component Analysis (PCA) (Jolliffe, 2011).

Comparing Eqs. (39) and (41) shows an interesting connection between the posterior of a class in QDA and the kernel over the the data instances of the class. In this comparison, the Eq. (41) should be considered for a class and not the entire data, so $\mathbf{K} \in \mathbb{R}^{n_k \times n_k}$, $\mathbf{D} \in \mathbb{R}^{n_k \times n_k}$, and $\mathbf{H} \in \mathbb{R}^{n_k \times n_k}$.

Now, consider the case where still the covariance matrices are all identity matrix but the priors are not equal. In this case, we have Eq. (38). If we take an exponential (inverse of logarithm) from this expression, the π_k becomes a scale factor (weight). This means that we still are using distance metric to measure the distance of an instance from the means of classes but we are scaling the distances by the priors of classes. If a class happens more, i.e., its prior is larger, it must have a larger posterior so we reduce the distance from the mean of its class. In other words, we move the decision boundary according to the prior of classes (see Fig. 2).